

# Moral Tensors and DecisionProofs

## Compiling Language into an Auditable, Grounded Safety Layer

Andrew H. Bond

San José State University · IEEE Senior Member

ORCID 0009-0003-2599-6158

Structure-preserving representation before decision contraction.

PyTorch Conference North America 2026 — Responsible AI poster

pip install erisml-compiler · github.com/ahb-sjsu/erisml-compiler · erisml-lib · agi-hpc

1

Don't collapse to a scalar.

|          | harm | rights | consent | risk | fair |
|----------|------|--------|---------|------|------|
| speaker  |      |        |         |      |      |
| village  |      |        |         |      |      |
| refugees |      |        |         |      |      |
| nazis    |      |        |         |      |      |

the moral tensor (who x what is owed)

✓ keep the structure

who is affected · what is owed  
who consented · who bears imposed risk

contract to a decision only at the end,  
and log the contraction:  
weights · who lost · what residue remains

★ 0.42

one scalar — structure lost

3

Four lenses. No silent winner.

one MoralGraph

Consequentialist  
harm/care tensor  
Gini · worst-off · Shapley

Deontic (Kantian)  
gates · universalizability  
via Z3 SMT

Virtue  
Aristotelian  
habit-consistency

Care ethics  
Gilligan · Noddings  
· Tronto

verdicts disagree? → cross\_projection\_disagreement

defers to a human

"We run four and show you where they conflict — we don't pick one and hide it."

5

Safety in the loop. Fails safe.

Agent planner

Reflex  
< 100 μs · hard stops

Tactical  
ErisML · 10-100 ms

Strategic  
policy + human oversight

ALLOW · REVISE · BLOCK

proof → proof → proof → proof

SHA-256 DecisionProof chain · previous\_proof\_hash → proof\_hash

ethics service times out → rule-based fallback. Never fails open.

2

pip install erisml-compiler — structure before the scalar

text

segment

extract

canonicalize

MoralGraph

tensorize

DEME

DecisionProof

MoralGraph nodes: stakeholder · act · maxim · commitment · fact · norm — carries a canonical SHA-256 hash

extractor tiers: rule (deterministic, silicon-castable) · LLM (NRP / vLLM + critic) · probe (LaBSE head)

alpha v0.9.0 · 477 tests · MIT · Zenodo DOI 10.5281/zenodo.20659432

4

Is this 'good'? → Does it preserve the bonds?

Obligation

Claim

L

N

s

r²

s = correlative swap (agent ↔ patient)

r² = deontic negation

Liberty

No-claim

s and r² commute → V4 (Klein four-group, order 4) — measured

full D4 (order 8) posited — quarter-turns not yet observed

0.0 perfect

0.155 baseline

0.25 warn

0.30 block

Bond Index: does the judgment survive the swap?

multi-rank tensors: party x time x action x coalition

V4 claim machine-checked: Lean 4 + Mathlib (formal/HohfeldV4.lean)

6

What the model says vs what it exhibits

transformer (Qwen2.5-7B, LLaMA, Mistral)

text lens (what it says)

activation lens (what it exhibits)

delta lens compare

requires human review (5 failure modes)

activation / probe lens is EARLY — uncalibrated by default (research-grade)

7

Auditable means you can replay the judgment

| stakeholder | harm | verdict |
|-------------|------|---------|
| speaker     | 0.76 | forbid  |
| village     | 0.83 | forbid  |
| refugees    | 0.00 | prefer  |
| nazis       | 0.18 | neutral |

Gini(harm) = 0.43 · worst-off = village

Shapley: speaker 7.11 · refugees 7.70  
nazis 7.88 · village 7.18

DecisionProof: proof\_hash → audit.ir\_hash

worked example examples/nazi\_attic · rank-2 DEME — one command, real numbers, a hash you can verify.

One command. Real numbers.  
A hash you can verify.

\$ erisml-compile examples/nazi\_attic

per-party verdicts · Gini 0.43 · exact Shapley  
DecisionProof chained to the IR hash — replay the judgment

- Structure before contraction — keep the tensor, log the collapse + the residue.
- Pluralism is the responsible move — four lenses, defer to a human on conflict.
- Auditable by construction; fails safe, never open.

alpha v0.9.0 · text path solid · activation lens early / uncalibrated  
ranks 1–3 real, higher partial · V4 measured, D4 posited (Lean + Mathlib verified)  
embodied = design target · conversational agents = live